Skill or Structure? Auditing Learned Skill and Explanation Faithfulness in Residual Reinforcement Learning for Portfolio Allocation

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ABSTRACT

Reinforcement learning (RL) allocators are often credited with reducing out-of-sample left-tail risk, yet long-only allocation already inherits downside protection for free from the simplex constraint, log-reward geometry, and the classical rebalancing premium. We ask how much of an RL allocator's apparent skill is genuinely learned rather than inherited structure, and whether post-hoc explanations of the trained policy identify its true decision mechanism. We train PPO as a *residual* on a structural-null base policy (equal-weight, volatility-scaled, or risk-parity) with a *structure-baselined* reward that credits only return earned above the base, and calibrate the resulting skill measure on a synthetic market with a toggleable predictive signal: skill rises monotonically with forecastability and vanishes when prediction is impossible. On real ETFs (2005–2024, 10 bps costs, walk-forward), both a scalar de-risking gate and a more expressive per-asset tilt add *no* skill above structure once measured net of a phase-randomization placebo null, across three bases and many seeds; the expressive agent loses more. We then probe the trained policy with per-feature causal ablation and compare it against gradient saliency and KernelSHAP. Attribution is broadly faithful to the causal mechanism, but which method is most faithful is agent-dependent, and SHAP recurrently over-credits a salient, high-variance feature. The synthetic method-agreement structure transfers to the real agent, whose decision rule is coherent and faithfully readable but not skillful.

CCS CONCEPTS

• **Computing methodologies** → **Reinforcement learning**; • **Theory of computation** → *Sequential decision making*.

KEYWORDS

reinforcement learning, portfolio allocation, explainability, attribution, causal intervention, tail risk, residual policy learning

1 INTRODUCTION

Model-free reinforcement learning has been reported to beat mean–variance optimization and reduce out-of-sample left-tail risk in portfolio allocation [9, 17]. Two long-standing critiques temper this claim. First, evaluations frequently omit transaction costs and rarely clear a strong $1/N$ benchmark [4, 8]. Second, the "skill" being measured may not be skill at all: a long-only allocator inherits downside protection mechanically from the probability-simplex constraint, from maximizing log wealth, and from the rebalancing premium of Stochastic Portfolio Theory [6]. A volatility-managed or risk-parity rule delivers much of the reported tail benefit with no forecasting [11, 12].

We treat these critiques as a measurement problem and ask two questions. **(RQ-skill)** How much of an RL allocator's apparent tail-risk benefit is learned skill above the free structure, rather than the structure itself? **(RQ-faithful)** When we explain the trained policy with standard attribution tools, do those explanations identify the mechanism the policy actually uses?

Our approach isolates skill by construction. We fix a *structural-null base policy* that already harvests the free structure, train the agent as a *residual* on that base, and reward it only for return earned *above* the base on the same market path. A zero residual reproduces the base exactly, so the zero-skill floor is free and the gradient never sees the structural growth. We calibrate this measure on a synthetic market whose predictive signal can be switched off, giving a ground-truth boundary where genuine skill is impossible. We apply the calibrated measure to real ETFs, and finally probe the trained policy's mechanism with causal interventions, adjudicating gradient saliency and KernelSHAP against what those interventions reveal.

Contributions. (1) A structure-baselined residual-RL skill measure, validated against synthetic ground truth where skill rises with forecastability and vanishes without it, for a scalar and an expressive agent. (2) A real-data verdict: neither a scalar de-risking gate nor an expressive per-asset tilt adds skill above structure net of a placebo null, across three bases and many seeds; the expressive agent loses *more*. (3) A causal-vs-attribution faithfulness probe across three agents, showing attribution is broadly faithful but its reliability is agent-dependent, with a recurring SHAP salience bias, and that the synthetic method-agreement transfers to the real agent. We frame success as a rigorous, honest verdict, not as beating the market.

2 RELATED WORK

RL allocators span value- and policy-based methods and are increasingly paired with interpretability claims [2, 3, 9]. The promise of tail-risk reduction is prominent in this line, but evaluations often use weak baselines and omit turnover costs. A parallel, skeptical thread revisits these results under realistic frictions and a strong naive-diversification benchmark, and frequently finds the advantage does not survive [4, 8]. Our residual formulation borrows residual policy learning from robotics, where a learned policy corrects a fixed controller [7, 15]; we repurpose it as a measurement instrument rather than a performance booster, pairing it with a structure-baselined reward and a synthetic ground-truth test that fixes where "no skill" lies. On explanation, gradient saliency [16] and KernelSHAP [10] are the standard post-hoc tools, and interpretable RL allocators increasingly report them [2, 3]; however, saliency maps for deep RL have been shown to be exploratory rather than explanatory [1]. We contribute a controlled, causally grounded test of what these tools recover for RL allocators specifically, adjudicated against interventional ground truth where it exists.

Our skill measure is complementary to benchmark-relative evaluation. Where prior work asks whether an RL allocator beats $1/N$ or mean–variance on net returns [4, 8], we ask a sharper question: does the agent add value *above the specific transparent rule it is built on top of*? Because the base is nested inside the policy (a zero residual is the base), the comparison is exact and per-path rather than across-strategy, which removes the market-path confound that makes cross-strategy Sharpe comparisons noisy and sample-dependent. This mirrors, on the evaluation side, what residual policy learning [15] does on the optimization side: define performance relative to a known-good reference so that only the increment must be learned and judged.

3 METHOD

3.1 Background: the free structure

Three properties give a long-only allocator downside protection with no forecasting. The probability-simplex constraint ($w \geq 0$, $\mathbf{1}^\top w = 1$) bounds any single position and forces diversification. Maximizing expected *log* wealth is implicitly variance-averse, penalizing the large drawdowns that compound destructively. And the rebalancing premium of Stochastic Portfolio Theory [6] shows that a diversified, periodically rebalanced portfolio earns an excess growth rate over its buy-and-hold counterpart purely from the geometry of rebalancing. A base policy that harvests these—equal-weight, volatility-scaled [12], or equal-risk-contribution risk parity [11]— therefore already reduces tail risk before any learning takes place. The empirical question is whether an RL residual adds anything *beyond* that free structure, and our design is built so that this increment is exactly what the reward measures.

3.2 Structure-baselined residual RL

Let $b_t \in \Delta^{n-1}$ be the weights of a fixed base policy on the probability simplex, computed from a trailing window: equal-weight, inverse-volatility (volatility-scaled), or equal-risk-contribution risk parity [11]. The agent acts as a residual on b_t in one of two modes.

De-risking gate. A scalar action $g_t \in [0, 1]$ shifts weight toward a safe asset s: $w_t = (1 - g_t)\, b_t + g_t\, e_s$. Here $g_t = 0$ reproduces the base and larger g_t times de-risking.

Expressive tilt. A per-asset action $a_t \in \mathbb{R}^n$ gives $w_t = \Pi_\Delta\big(b_t + \tau \tanh(a_t)\big)$, where Π_Δ is Euclidean projection onto the simplex [5] and τ bounds the tilt. A zero action again reproduces the base, but the agent can express any long-only deviation. An early *unbounded* per-asset residual (no tanh/τ bound) over-tilted and produced negative skill even synthetically; bounding the tilt fixed this and motivated reporting both the parsimonious gate and the bounded expressive tilt.

The reward is *structure-baselined*: the agent's net log return minus the base's net log return on the same path, each charged its own turnover cost,

$$r_t = \log\big(1 + w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1\big) - \log\big(1 + b_t^\top \rho_t - c\|b_t - b_{t-1}\|_1\big), \tag{1}$$

where ρ_t are asset returns and c the per-unit cost. The cumulative reward is realized *skill above structure*; it is zero for the base itself, so the gradient sees only what the agent adds. We train PPO [14] with an MLP policy via Stable-Baselines3 [13]. The observation is a trailing return window plus realized volatility, and (for the tilt) dual-horizon volatility, momentum, and the base weights, with a de-risking signal.

3.3 Synthetic ground truth

To calibrate the measure we generate regime-switching markets with a *toggleable* leading signal of strength $\sigma \in [0, 1]$: a two-asset risky/safe world for the gate and a multi-asset world for the tilt. When σ is high, timed de-risking is possible; when $\sigma = 0$ the market is unforecastable and genuine skill is impossible by construction. A valid measure must rise with σ and vanish at $\sigma = 0$.

3.4 Real-data placebo null

On real data there is no ground truth for "no skill." We report skill *net of a phase-randomization placebo null*: surrogate series that preserve each asset's power spectrum but destroy volatility clustering, retrained identically. The placebo mean is the luck/overfitting floor; a positive floor is expected and is exactly why netting is necessary. We report skill net of this null with a small-sample t confidence interval and the fraction of placebo draws that match or beat the real agent (placebo exceedance). Agents are evaluated with an anchored expanding-window walk-forward and per-fold retraining, and placed beside $1/N$, minimum-variance, and CVaR-minimizing baselines on the same out-of-sample window.

3.5 Causal-vs-attribution faithfulness probe

Given a trained policy we replay its deterministic decisions and measure two tracks over the observation features. The *causal* track ablates each feature (freeze to its mean; permute across time) and records the mean absolute change in the policy's scalar decision, a direct measure of what the policy responds to; we also inject volatility shocks and flip the signal as legibility checks. The *attribution* track computes gradient saliency [16] and KernelSHAP [10].

Two confounds must be removed before the tracks are comparable. *Scale*: the signal feature has 35–100× the input scale of the return and volatility features, and raw gradient saliency $|\partial g/\partial o_i|$ is scale-invariant whereas ablation and SHAP are scale-sensitive, so the three tracks otherwise live in different units and a disagreement could *falsely* read as unfaithfulness; we place saliency in gradient×standard-deviation units to match. *Cardinality*: the return block has many features (100 in the tilt observation, 220 in the real gate observation) versus a single signal feature, so summed group importance over-weights the large group. We therefore (i) express every method as normalized per-feature shares and (ii) compute the causal group verdict from per-feature (not joint) ablation, so causal and attribution share one basis. Normalization is a monotone rescale and does not by itself cancel a group's feature-count advantage; the cardinality-robust statistic we report is the per-feature Spearman correlation between the causal profile and each attribution profile.

Because normalized shares always sum to one, they hide an inactive agent. We therefore also report *activity diagnostics*: the standard deviation of the policy's scalar decision over the rollout, and the raw, un-normalized |causal effect| per feature group. If importances collapse to zero the probe records a logged "no measurable

Table 1: Synthetic validation: mean structure-baselined skill (held-out, 5 seeds). Skill rises with forecastability and vanishes at the null.

Signal strength σ (gate)	Mean skill
0.00 (unforecastable null)	5.9×10^{-5}
0.50	1.7×10^{-4}
0.95 (strongly forecastable)	2.9×10^{-4}
Expressive tilt, net-of-null (signal 0.95):	
skill off / skill on	-6.4×10^{-5} / $+1.4 \times 10^{-4}$
skill net of churn floor	$+2.0 \times 10^{-4}$ (4.1×)

mechanism" null rather than dividing by zero. On synthetic data the leading signal is the known driver; on real data no oracle exists, so only *method agreement* is adjudicable.

4 EXPERIMENTS

4.1 Setup

The real universe is 11 liquid ETFs spanning equity sectors, Treasuries, gold, and international equity (2005–2024, daily, 10 bps costs). The gate observation is [window×n returns | n short vol | 1 signal] (232-dim for the real panel with window 20); the tilt observation adds dual-horizon volatility, momentum, and the base weights (121-dim in the 5-asset synthetic world). We use PPO with an MLP policy, 150k training steps, and five seeds; real robustness sweeps use five agent seeds and ten placebo draws per base. The turnover cost is $c\|w_t - w_{t-1}\|_1$ at 10 bps, charged identically to agent and base. The walk-forward uses a 1260-day (five-year) initial training window and 252-day (one-year) test blocks, retraining per fold and stitching the held-out predictions; baselines are rolled over the same out-of-sample calendar. On real data the safe sleeve is resolved by ticker (intermediate then long Treasuries), never a hard-coded index, because vendor columns are alphabetical. Heavy training runs on a CPU batch cluster with thread caps; a GPU gives no benefit for this small policy, whose bottleneck is environment stepping rather than matrix multiplication. All reported figures are held out.

4.2 The skill measure is valid (RQ-skill, synthetic)

Table 1 and Figure 1 show mean skill rising monotonically with signal strength for the gate and vanishing toward the unforecastable null. The expressive tilt cannot reach a costless do-nothing floor—with no signal it over-churns and loses about -6.4×10^{-5}—so it is judged net of that churn null: skill-on minus skill-off is $+2.0 \times 10^{-4}$, about four times the floor. The measure detects learned skill only when learned skill is possible, for both agents. A caveat we carry forward: at higher training budgets the signal-off floor is noisy and slightly positive, because PPO extracts small *spurious* skill from pure noise; that floor is the overfitting baseline and is why the real-data analysis nets against a matched null (§4.3).

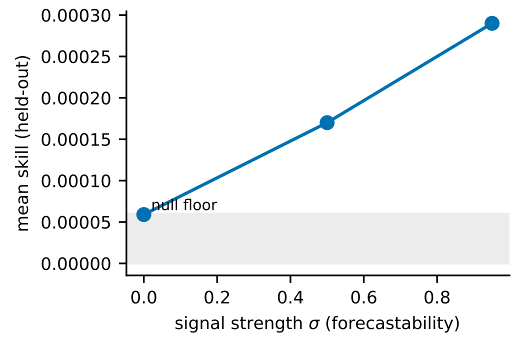

Figure 1: Synthetic skill vs. signal strength (gate). Mean held-out skill rises monotonically with forecastability and collapses toward the noisy null floor (shaded band) at $\sigma = 0$, exactly the behaviour a valid skill measure must show.

Table 2: Real ETFs, *gate*: skill net of the placebo null (5 agent seeds, 10 placebo draws per base). All intervals below zero; exceedance = 1.00.

Base	Agent	Placebo	Skill net	95% CI ($\times 10^{-4}$)
Equal-weight	−8.6e-5	+9.0e-5	−1.76e-4	[−2.9, −0.7]
Vol-scaled	−7.1e-5	+4.8e-5	−1.19e-4	[−2.1, −0.3]
Risk-parity	−6.0e-5	+4.1e-5	−1.01e-4	[−1.4, −0.6]

4.3 No skill above structure (RQ-skill, real ETFs)

Tables 2 and 3 report skill net of the placebo null across the three bases, for the gate and the tilt, and Figure 2 summarizes them. Every skill-net confidence interval lies entirely below zero, the placebo (luck) mean is positive everywhere (so netting is essential), and every structureless surrogate beats the real agent (placebo exceedance = 1.00) for all bases and both agents. The gate learns to mostly do nothing (mean gate $\approx$ 0.04), tracking its base on every reported tail and risk metric—expected shortfall, maximum drawdown, tail ratio, skewness, Sharpe, and Sortino—while both the agent and its base clearly beat naive $1/N$ on the downside. The agent-versus-base differences are within noise, so the tail benefit over $1/N$ is inherited structure (risk-parity / de-risking geometry), not learned skill: the agent does not improve on the base it starts from. Skill-net is most negative for the weakest base (equal-weight), where the gate activates most and churns without edge. The expressive tilt loses *more* than the gate (1.5–1.8×) in the same base ordering: more expressiveness means more churn, and where no timeable structure exists that churn is pure cost drag. A capable agent sharpens the verdict rather than rescuing it.

4.4 Is the explanation faithful? (RQ-faithful)

Table 4 summarizes the probe across three agents. On the synthetic gate, where the signal is the known driver, KernelSHAP names it the top feature in all five seeds and gradient saliency in four of five—the miss is the least signal-concentrated agent (causal signal

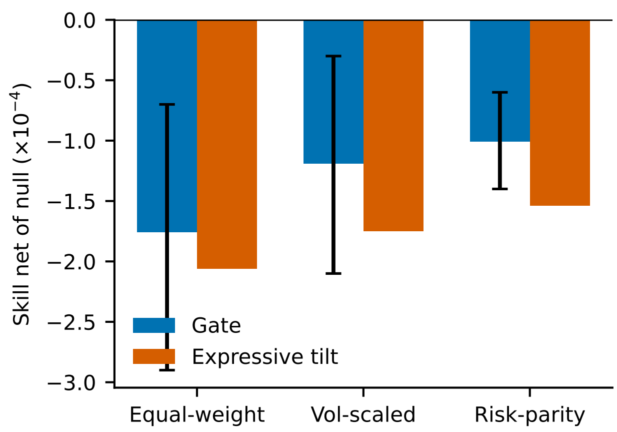

Figure 2: Skill net of the placebo null on real ETFs, by structural base. Bars lie below zero for every base and both agents (gate error bars are 95% CIs); the expressive tilt is strictly worse than the gate. Placebo exceedance is 1.00 **throughout.**

Table 3: Real ETFs, *expressive tilt*: skill net of the null. Same sweep; all intervals below zero, exceedance = 1.00**. Loses more than the gate.**

Base	Agent	Skill net	vs. gate
Equal-weight	−1.56e-4	−2.06e-4	1.17×
Vol-scaled	−1.26e-4	−1.75e-4	1.47×
Risk-parity	−1.03e-4	−1.54e-4	1.52×

share 0.54 vs. $\approx$ 1.0)—and per-feature agreement with the causal profile is positive for both.

On the synthetic expressive tilt, saliency is near-perfectly aligned with the causal profile (+0.999) while SHAP over-credits the salient, high-variance signal feature (+0.88), naming it the top group in four of five seeds when the causal analysis does not; this inverts which method is more reliable relative to the gate. The cardinality caveat bites here: with 100 return features versus 1 signal, the group-level "top driver" verdict is confounded—per feature the signal is actually the strongest single feature ($\sim$ 54× an average return feature)—so the per-feature Spearman is the honest measure.

On the real agent there is no ground truth, so we report method agreement only. The activity diagnostic overturns a natural prior: although the mean gate is $\approx$ 0.05, its standard deviation is $\approx$ 0.18, so the agent actively swings the gate rather than sitting still. It has a real, if unprofitable, decision mechanism (none of five seeds degenerate). Given that live mechanism, per-feature agreement with causal ablation is strong—saliency +0.996, SHAP +0.79— preserving the saliency-over-SHAP ordering seen synthetically. All four methods attribute about 0.96 of the normalized share to the recent-returns block (causal freeze 0.963, permute 0.961; saliency 0.962; SHAP 0.922; Figure 3), with the crisis signal a minor contributor; SHAP again shifts slightly more mass onto the signal (0.05 vs. causal 0.014), the same salience bias. Unlike the synthetic tilt this is not a cardinality artifact, since the signal is only about three times an

Table 4: Faithfulness probe: per-feature Spearman between the causal profile and each attribution method (5 seeds). Real-data numbers are method agreement, not faithfulness (no ground-truth driver).

Agent	Saliency	SHAP	Note
Synthetic gate	+0.79	+0.61	SHAP names driver 5/5; saliency 4/5
Synthetic tilt	+0.999	+0.88	SHAP over-credits salient signal
Real gate	+0.996	+0.79	agreement only; agent is active

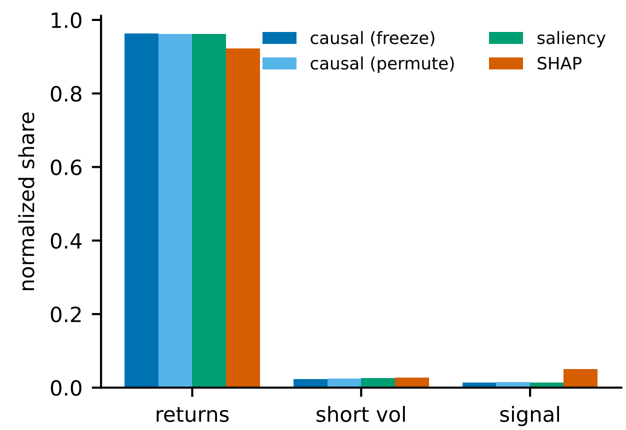

Figure 3: Real agent: normalized per-feature group shares by method. All four tracks agree the recent-returns block dominates the de-risking decision; SHAP places slightly more mass on the salient signal than the causal tracks do, the same bias seen synthetically.

average return feature. The real allocator has a coherent, faithfully readable decision rule; the rule simply is not skillful.

As a directional legibility check, the interventions behave as a de-risking policy should: injecting a transient volatility shock into the replayed path raises the gate (more weight to the safe sleeve) relative to the unshocked path, and forcing the crisis signal high does likewise. The agent thus reads volatility and the signal in the expected direction; what it lacks is not responsiveness but an edge that survives costs.

5 DISCUSSION AND LIMITATIONS

The questions close a loop. The measure detects skill when it exists (synthetic), finds none on real markets net of luck (both agents, three bases), and shows the trained policy's mechanism is largely readable by post-hoc attribution, with saliency tracking the causal per-feature profile more tightly than SHAP across all three agents. The recurring practical failure mode is SHAP over-crediting a salient, high-variance feature; which tool to trust is agent-dependent, so an interpretability report for an RL allocator should cross-check attribution against a cheap causal ablation rather than trust a single method. The real-agent result is a useful corrective on its own: a low mean action can hide an active decision rule, and only an absolute-magnitude diagnostic—not a normalized share—reveals whether there is a mechanism to explain at all.

Two implications for practice follow. First, an interpretability report for an RL allocator should not lean on a single attribution method: because reliability is agent-dependent, a cheap causal ablation is worth running as an arbiter, and a large saliency–SHAP disagreement is a signal to do so. Second, the result that the *more* expressive agent loses *more* is not incidental. With no timeable edge in the data, every additional degree of freedom the policy can move is an additional way to incur turnover cost without compensating return; expressiveness is therefore a liability, not an asset, precisely when there is no skill to express. This argues for reporting RL-allocator results against a structural base at several expressiveness levels, as we do, rather than at a single architecture.

Several limitations bound these claims. The synthetic market is deliberately simple, chosen so that "no skill" is well defined rather than realistic. The real study uses one liquid universe and daily data; other universes, frequencies, or richer action spaces may differ. Attribution on real data measures agreement, not faithfulness, because no oracle mechanism exists; high agreement cannot rule out two methods being wrong together, though causal ablation is the most direct available reference. The per-feature causal vectors were aggregated to groups rather than persisted in full; a camera-ready extension should save them to state per-feature dominance directly. Finally, the placebo null destroys volatility clustering but preserves linear autocorrelation; alternative nulls are worth reporting.

6 CONCLUSION

Treating RL portfolio allocation as a measurement problem, we separate learned skill from inherited structure with a structure-baselined residual and a synthetic ground-truth calibration, and test whether standard explanations identify the trained policy's mechanism. On real ETFs, neither a parsimonious nor an expressive RL allocator earns skill above a transparent structural base once costs and a luck null are accounted for. The policy's decision rule is nonetheless coherent and faithfully readable, most reliably by gradient saliency; KernelSHAP carries a salience bias. Code and configurations will be released publicly.

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